

ASHOK KUMAR REDDY KOTA

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PROFESSIONAL SUMMARY:

Backend-Focused Full-Stack Software Engineer with 2.5+ years of experience building scalable SaaS platforms, enterprise applications, educational management systems, and AI-powered backend services. Strong expertise in Python, FastAPI, Django, REST APIs, asynchronous architectures, database optimization, and distributed system design. Experienced in developing and maintaining government-sector educational platforms, FinTech-grade ledger systems, and multi-tenant applications. Proven ability to deliver secure, high-performance software solutions while integrating modern cloud and AI technologies. Passionate about scalable architecture, performance optimization, and production-grade backend engineering.

TECHNICAL SKILLS:

Backend & APIs: Python, FastAPI, Django, Django REST Framework, Flask, RESTful APIs, Async Programming, Background Tasks, API Design, API Versioning

Authentication & Security: OAuth 2.0, OpenID Connect, Keycloak, AWS Cognito, JWT, RBAC, Password Hashing, Token Rotation, TLS 1.3, AES-256

Frontend: React.js, TypeScript, JavaScript, Redux Toolkit, RTK Query, HTML5, CSS3, Real-Time UI Development

System Design: Microservices, Event-Driven Architecture, Scalable Architectures, Distributed Systems, Multi-Tenant SaaS, Design Patterns

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Schema Design, Query Optimization, Database Management, Indexing, Caching, Audit Logs

Cloud & AI: AWS (EC2, S3, RDS, CloudFront, Cognito), AWS Bedrock, Hugging Face, Llama 3 70B, Qwen3, Prompt Engineering, RAG, AI Microservices

DevOps & Testing: Docker, Git, GitHub, CI/CD Concepts, Pytest, Unit Testing, Integration Testing, Postman, Logging & Monitoring

PROFESSIONAL EXPERIENCE:

AppZone Technology — Software Engineer

Hyderabad, India | Mar 2026 – Present

Client: Andhra Pradesh Tribal Welfare Residential Educational Institutions Society (APTWREIS)

Project: AP Gurukulam Educational Management Platform

- Develop and maintain enterprise-scale educational management applications using Python, Django, REST APIs, and MySQL, supporting students, teachers, administrators, and government stakeholders.
- Design, develop, and enhance modules, workflows, dashboards, and services while implementing new business requirements and improving existing functionalities.
- Manage and optimize MySQL databases through schema design, query tuning, indexing, data validation, and performance improvements to ensure system scalability and reliability.
- Implement role-based access control (RBAC), troubleshoot production issues, resolve application defects, and maintain the stability and security of live production systems.
- Collaborate with stakeholders to gather requirements, deliver software enhancements, modernize legacy applications, and contribute to deployment, monitoring, and ongoing platform maintenance.

CosmoOps Private Limited — Software Engineer

Pondicherry, India | Dec 2023 – Feb 2026

- Designed and deployed scalable FastAPI and Django REST services powering FinTech, AI-driven, and multi-tenant SaaS platforms.
- Engineered ledger-based financial systems with atomic, idempotent transactions and event-driven payment processing using Stripe and Razorpay webhooks.
- Built real-time systems using WebSockets, Redis pub/sub, and asynchronous background tasks to support high-concurrency workloads.
- Integrated LLM-powered automation using Hugging Face and AWS Bedrock for intelligent task prioritization, summarization, and analytics workflows.
- Optimized PostgreSQL and Redis through indexing and caching strategies, reducing API latency by up to 30% while enforcing OAuth2/OIDC-based RBAC security.

PROJECTS:

AP Gurukulam Educational Management Platform

Technologies: Python, Django REST Framework (DRF), REST APIs, HTML5, CSS3, Bootstrap, jQuery, Git

- Developed and maintained a large-scale educational management platform supporting students, teachers, administrators, and government stakeholders across tribal welfare institutions.
- Designed and implemented new modules, workflows, dashboards, reports, and backend services using Django, DRF, and REST APIs to streamline academic and administrative processes.
- Optimized SQL databases through query tuning, indexing, schema enhancements, data migration, and validation techniques to improve system performance and reliability.
- Implemented secure role-based access control (RBAC), resolved production issues, managed deployments, and contributed to continuous platform enhancements and modernization initiatives.

AcuFinAI: AI-Powered Digital Wallet & Smart Spending Engine

Technologies: FastAPI, React, TypeScript, Redux Toolkit, Celery, Redis, PostgreSQL, OpenSearch, AWS

Bedrock, Keycloak

- Built a double-entry, multi-currency ledger engine with atomic, idempotent transactions ensuring financial consistency and audit reliability.
- Designed event-driven payment workflows integrating Stripe and Razorpay with retry-safe reconciliation mechanisms.
- Developed AI-powered financial intelligence pipelines for expense categorization, anomaly detection, and personalized spending insights.
- Architected a secure, multi-tenant FinTech SaaS platform with OAuth2/OIDC-based RBAC and tenant isolation.

Collabryta: AI-Powered Remote Team Collaboration Platform

Technologies: FastAPI, React.js, TypeScript, WebSockets, Redis, Hugging Face, Qwen3, AWS

- Architected a scalable, multi-tenant collaboration platform enabling real-time messaging, task orchestration, and workflow automation.
- Engineered modular FastAPI microservices with async processing and Redis-backed pub/sub architecture.
- Integrated Llama 3 70B and Hugging Face Inference Endpoints for task prioritization, meeting summarization, and document intelligence.
- Implemented secure authentication, RBAC, Amazon Cognito integration, Amazon S3 storage, and CloudFront content delivery.

EDUCATION:

Bachelor of Technology (B.Tech) – Electrical & Electronics Engineering

Rajeev Gandhi Memorial College of Engineering & Technology 2019 – 2023

CERTIFICATIONS

- AWS Certified Developer – Associate | Docker Certification | GitHub Foundations
- Python Programming Essentials | FastAPI Development and REST API Design | Full-Stack Web Development with Django
- PostgreSQL Certification | MongoDB Certification | SQL and Relational Databases
- Redis Fundamentals | Microservices Architecture Fundamentals
- React | JavaScript and TypeScript | Introduction to Cloud DevOps
- AWS Cloud Practitioner Essentials | API Testing with Postman

ADDITIONAL HIGHLIGHTS

- 2.5+ years of professional experience in Backend and Full-Stack Software Development.
- Strong expertise in Python, FastAPI, Django, PostgreSQL, Redis, and AWS Cloud Services.
- Hands-on experience building FinTech platforms, Educational Management Systems, AI-powered applications, and SaaS products.
- Experienced in database design, performance optimization, production support, and application modernization.
- Strong foundation in distributed systems, system design, security, scalability, and cloud-native architectures.